Speaker Notes

Efficient Bayesian Inference for Multivariate Factor Stochastic Volatility Models

Kastner, Frühwirth-Schnatter & Lopes (2017)

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Running example throughout the entire talk. You manage a **global macro hedge fund** with positions in 26 currency pairs. Every morning you face one question: *what is today's risk landscape?* Which currencies move together? Where is volatility clustering? Which relationships have broken down? Use this example concretely on every slide — do not just mention it once.

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1. Motivation — From Univariate to Multivariate Stochastic Volatility

[SLIDE: Slide 3]

The core problem in one sentence

Managing 26 currencies means tracking a 26×26 covariance matrix — that is $\frac{26 \cdot 27}{2} = 351$ unique parameters, **every single trading day**. Estimating 351 numbers reliably from a single day's data is statistically hopeless.

Connecting to Task 2

In Task 2 we modelled a **single** return series — daily CHF/USD — by augmenting with latent weights ω_i to handle t_ν-errors. Each observation got its own effective variance σ^2/ω_i. That was a univariate SV model. The natural next step is: what if we have *many* assets simultaneously? We need not only time-varying variances but also time-varying **covariances**. The full covariance matrix has $\frac{m(m+1)}{2}$ free elements — growing quadratically in m.

The paper's answer: a factor structure

Rather than estimating 351 numbers directly, the paper imposes a **low-dimensional factor structure**: $r \ll m$ latent factors, each with its own SV process. Think of it as a financial autoencoder — you compress the 351-dimensional covariance structure into a r-dimensional bottleneck. Crucially, the bottleneck itself *breathes* dynamically over time.

> **Likely question:** "Is this just PCA?"
> Answer: PCA gives a *static* low-rank approximation with fixed loadings. Here the latent factors have *time-varying volatility* — their own stochastic processes. The covariance structure can change its shape every day, not just its scale. During the 2008 crisis, correlations spiked and the global dollar factor suddenly dominated everything — a static PCA cannot capture that.

2. The Factor SV Model

[SLIDE: Slide 4]

Observation equation

The model decomposes m-dimensional returns as:

$$\underbrace{\mathbf{y}_t}_{m\times 1} = \underbrace{\mathbf{\Lambda}}_{m\times r}\underbrace{\mathbf{f}_t}_{r\times 1} + \underbrace{\boldsymbol{\varepsilon}_t}_{m\times 1}, \qquad \mathbf{f}_t \mid \mathbf{h}_t \sim \mathcal{N}_r(\mathbf{0}, V_t), \qquad \boldsymbol{\varepsilon}_t \mid \mathbf{h}_t \sim \mathcal{N}_m(\mathbf{0}, U_t)$$

Three economic building blocks:

$\mathbf{\Lambda}$ — the loading matrix. Each row is one currency's *sensitivity profile*. AUD has a high loading on the commodity factor. JPY loads *negatively* on the risk-sentiment factor — it appreciates when other currencies fall. EUR loads on the global dollar factor. These loadings are unknown and must be inferred from data.

$\mathbf{f}_t$ — **the latent factors.** These are the **unobserved global forces** driving the system. Unlike Fama–French, you cannot look them up — they are inferred from the data. Each factor has its own stochastic volatility process, its own "currency VIX", evolving as an AR(1) in log-variance space.

$\boldsymbol{\varepsilon}_t$ — **idiosyncratic shocks.** What remains after removing factor contributions. The CHF spike of January 2015 — when the Swiss National Bank suddenly unpegged from the EUR — is a perfect illustration. No global factor caused that move. It is absorbed entirely into the idiosyncratic SV of CHF, leaving the factor structure undistorted.

The log-volatility state vector $\mathbf{h}_t$

$\mathbf{h}_t$ collects **all** $m + r$ **log-variances** at time t:

- $h_{1t}, \ldots, h_{mt}$: log-variances of the idiosyncratic shocks for each of the m currencies.
- $h_{m+1,t}, \ldots, h_{m+r,t}$: log-variances of each of the r factors.

Taking the *log* keeps variances positive without imposing any constraints. Each component evolves as an independent AR(1):

$$h_{it} \;=\; \mu_i \;+\; \phi_i(h_{i,t-1} - \mu_i) \;+\; \sigma_i\,\eta_{it}, \qquad \eta_{it} \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$$

μ_i is the long-run mean of the log-variance (i.e. the typical volatility level), $\phi_i \in (-1, 1)$ controls persistence (how quickly volatility mean-reverts), and σ_i is the vol-of-vol.

The time-varying covariance matrix

Given $\mathbf{h}_t$, the covariance matrix of returns is:

$$\operatorname{cov}(\mathbf{y}_t \mid \mathbf{h}_t) \;=\; \boldsymbol{\Lambda}\, V_t\, \boldsymbol{\Lambda}' \;+\; U_t$$

where $U_t = \operatorname{diag}(e^{h_{1t}}, \ldots, e^{h_{mt}})$ and $V_t = \operatorname{diag}(e^{h_{m+1,t}}, \ldots, e^{h_{m+r,t}})$.

When the global dollar factor's log-variance $h_{m+1,t}$ is large, V_t blows up, $\boldsymbol{\Lambda} V_t \boldsymbol{\Lambda}'$ blows up, and *all* pairwise correlations driven by that factor spike simultaneously. This is exactly the empirical phenomenon of **correlation breakdown during crises** that killed many pre-2008 risk models.

> **Likely question:** "Why are the factors uncorrelated?"
> V_t is *diagonal* — the factors are independent of each other *conditional on* $\mathbf{h}_t$. All cross-asset dependence is captured through the shared loading matrix $\boldsymbol{\Lambda}$, not through correlation between the factors themselves. This is the classical factor model assumption and is what makes the Gibbs sampler tractable.

> **Likely question:** "What is the difference between U_t and V_t?"
> U_t captures volatility that is *specific* to one currency — the CHF blow-up affected only CHF. V_t captures volatility that is *common* — a spike in the global dollar factor simultaneously moves all dollar-correlated currencies. The additive decomposition $\boldsymbol{\Lambda} V_t \boldsymbol{\Lambda}' + U_t$ separates these two channels cleanly.

3. Identification

[SLIDE: Slide 5]

The scaling ambiguity

The factor model has a fundamental identification problem. For any scalar $c \neq 0$:

$$\mathbf{\Lambda}\,\mathbf{f}_t \;=\; \underbrace{(c\,\mathbf{\Lambda})}_{\mathbf{\Lambda}^*}\underbrace{(c^{-1}\mathbf{f}_t)}_{\mathbf{f}_t^*}$$

The data observes only the product $\mathbf{\Lambda}\mathbf{f}_t$. It cannot distinguish a loading of 2 with a factor of 1 from a loading of 1 with a factor of 2. The posterior has a **continuous ridge of equally valid solutions** — MCMC will wander along this ridge indefinitely without additional constraints.

The classical fix and its problem

The classical approach pins $\lambda_{jj} = 1$ for each factor — forcing, say, EUR/USD to be the "leader" of factor 1, GBP/USD the leader of factor 2, and so on. This works mechanically but is **economically arbitrary**. Why should EUR/USD be privileged over GBP/USD? The ordering imposes a hierarchy the data did not ask for.

The paper's approach

Instead of constraining $\mathbf{\Lambda}$, the paper constrains the **long-run mean of each factor's SV process** by fixing:

$$\mu_{m+j} = 0 \qquad \Longrightarrow \qquad \mathbb{E}\big[e^{h_{m+j,t}}\big] \approx 1.$$

The factor's variance is anchored to 1 *on average over time*. It can still be high during crises and low during calm periods — the stochastic volatility is alive — but it cannot drift to an arbitrary permanent scale. Crucially, λ_{jj} is left *unrestricted*.

> **Why this matters for MCMC.** Unresolved non-identifiability creates a multimodal posterior — MCMC chains get stuck between modes. But the paper's approach leaves a *residual* sign ambiguity in λ_{jj} (positive or negative) which becomes a **computational tool**: the ASIS algorithm will exploit this freedom to make much larger jumps in parameter space. Stress this connection to Slides 9–10.

> **Likely question:** "Doesn't fixing $\mu_{m+j} = 0$ lose information?"
> No. It only fixes the *scale* of the factor — the information about scale is simply redistributed into λ_{jj}. The model is observationally equivalent to the classical fix; it is a different choice of where to absorb the scaling ambiguity.

4. Bayesian Inference — Priors & Data Augmentation

[SLIDE: Slide 6]

Priors — encoding what we know before seeing data

Priors are not merely "starting points" for the algorithm. They encode **beliefs about plausible parameter values before observing any data**, and they influence the entire posterior. A useful metaphor: the prior is a gravitational field pulling the sampler toward plausible regions; the likelihood is another force pulling toward regions that fit the data; the posterior is the equilibrium.

The priors are set independently for each SV process $i = 1, \ldots, m + r$:

$$\mu_i \sim \mathcal{N}(b_\mu,\, B_\mu), \qquad \frac{\phi_i + 1}{2} \sim \mathcal{B}(a_0,\, b_0), \qquad \sigma_i^2 \sim B_\sigma \cdot \chi_1^2$$

$\mu_i \sim \mathcal{N}(b_\mu, B_\mu)$ — **long-run log-variance level.** μ_i lives on the whole real line (log-variance is unrestricted), so a Normal prior is natural and conjugate. For daily currency returns, typical annualised volatility is around 8–15%, so daily variance is roughly $(0.10)^2/252 \approx 4 \times 10^{-5}$, giving $\mu_i \approx \log(4 \times 10^{-5}) \approx -10$. The hyperparameters b_μ and B_μ encode your prior guess at the typical volatility level and your uncertainty about that guess.

$(\phi_i + 1)/2 \sim \mathcal{B}(a_0, b_0)$ — **persistence.** $\phi_i \in (-1, 1)$ is the AR(1) persistence coefficient. The **reparameterisation trick**: adding 1 and dividing by 2 maps $(-1, 1)$ onto $(0, 1)$, where the Beta distribution is defined. This enforces stationarity automatically — no truncation needed.

The Beta distribution is extremely flexible. The key insight is that $\mathcal{B}(a_0, b_0)$ **with** $a_0 \gg b_0$ **puts most mass near 1**, encoding the empirical fact of *volatility clustering*: the VIX stays elevated for months after a crisis, it does not reset overnight. You are not forcing persistence — you are expressing a well-founded prior belief while remaining open to the data overriding it. The default in the `stochvol` package is $\mathcal{B}(5, 1.5)$.

$\sigma_i^2 \sim B_\sigma \cdot \chi_1^2$ — **vol-of-vol.** σ_i controls how wildly log-volatility itself fluctuates. The χ_1^2 distribution is simply the distribution of Z^2 for $Z \sim \mathcal{N}(0, 1)$ — it puts most mass near zero and has a long right tail. This reflects the belief that log-volatility usually drifts smoothly (small σ_i) but can occasionally jump (heavy tail). The scaling constant B_σ sets the overall plausible magnitude.

$\lambda_{ij} \sim \mathcal{N}(0, B_\lambda)$ — **factor loadings.** Ridge-type shrinkage. Most currencies probably do not load heavily on most factors. The Gaussian prior centred at zero pulls weak loadings toward zero unless the data insists otherwise. Bayesian Ridge regression, expressed in prior form.

Data augmentation — the computational trick

Treat the log-volatility paths $\mathbf{h} = \{h_{i,\bullet}\}$ and the factor paths $\mathbf{f} = \{\mathbf{f}_t\}$ as **missing data**. This breaks the intractable joint estimation problem into two simple conditional pieces:

- **Conditional on f and $\boldsymbol{\Lambda}$:** Each of the $m+r$ residual series is an independent univariate SV model $\Rightarrow$ hand to `stochvol`.
- **Conditional on h:** $\boldsymbol{\Lambda}$ and $\mathbf{f}_t$ have closed-form Normal posteriors $\Rightarrow$ standard weighted regression updates.

Direct parallel to Task 2. Augmenting with ω_i turned an intractable t-likelihood into a conditionally Normal–Gamma system. Exactly the same philosophy: introduce a latent variable that makes the conditional distributions tractable, then alternate between sampling the latent variable and the parameters. This is the Gibbs sampler in its purest form.

Likely question: "Why not just integrate out $\mathbf{f}$ and $\mathbf{h}$ analytically?"
Because the SV log-variance equation is nonlinear — the $e^{h_{it}}$ terms make direct marginalisation intractable. Data augmentation sidesteps this by conditioning on the nonlinear terms, turning everything into linear Gaussian problems.

5. The Standard Gibbs Sampler

[SLIDE: Slide 7]

The Gibbs principle

Even when the joint posterior $p(\mathbf{\Lambda}, \mathbf{f}, \mathbf{h} \mid \mathbf{y})$ is intractable, the **conditional** posteriors are simple. The Gibbs sampler alternates: fix everything except one block, sample that block from its conditional, move to the next block, repeat. Each new draw depends only on the current state — not on the full history. This is the Markov chain property. Over many iterations the chain converges to the true joint posterior as its stationary distribution.

Step (a) — Univariate SV updates ($m + r$ independent problems)

Conditional on $\mathbf{f}$ and $\mathbf{\Lambda}$, subtract the factor component from each currency's returns:

$$\tilde{y}_{it} \;=\; y_{it} - (\mathbf{\Lambda}\mathbf{f}_t)_i.$$

What remains is a pure idiosyncratic SV series for currency i. Similarly, each factor series f_{jt} is driven purely by its own SV process. You now have $m + r$ independent univariate SV problems. Update $(h_{i,\bullet},\, \mu_i,\, \phi_i,\, \sigma_i)$ for each via `stochvol` using the log-squared observation trick.

Why "stochastic"? Log-volatility h_{it} is a *latent random variable* with an independent shock η_{it} hitting it each period. It is not a deterministic function of past returns (as in GARCH). Volatility is genuinely unobserved and randomly evolving — hence stochastic volatility.

Finance illustration. After removing the global dollar factor's contribution from EUR/USD returns, the residual is a univariate SV series with its own persistence, long-run level, and vol-of-vol. The SNB unpegging in January 2015 appears as a massive spike in CHF's idiosyncratic log-variance — visible only in that series, not in the factor volatilities.

Step (b) — Sample loadings $\mathbf{\Lambda}$

Conditional on $\mathbf{f}$ and $\mathbf{h}$, the observation equation is a **weighted Normal linear regression** for each row of $\mathbf{\Lambda}$:

$$y_{it} = \boldsymbol{\lambda}_i' \mathbf{f}_t + \varepsilon_{it}, \qquad \varepsilon_{it} \sim \mathcal{N}(0,\, e^{h_{it}}).$$

Each observation t is weighted by $e^{-h_{it}}$ — calm periods (low variance) get upweighted, crisis periods (high variance) get downweighted. The posterior for each row is a multivariate Normal with a closed-form expression. m independent regression problems, each with T observations.

Finance illustration. EUR/USD's loading on the dollar factor is estimated most reliably from quiet trading days, not from crisis spikes when idiosyncratic noise dominates. The weighting scheme implements this automatically.

Step (c) — Sample factors $\mathbf{f}_t$

Conditional on $\mathbf{\Lambda}$ and $\mathbf{h}$, each $\mathbf{f}_t$ has a **multivariate Normal posterior** — T independent r-dimensional updates. Given that EUR/USD loads 0.8 on the dollar factor, today's EUR/USD return pins down what the dollar factor probably was today.

The punchline

> **"In principle correct — in practice broken."** The sampler is theoretically valid and will eventually converge to the correct posterior. But the strong posterior correlation between $\mathbf{\Lambda}$ and $\mathbf{f}_t$ means the chain moves so slowly that "eventually" requires millions of iterations beyond what is computationally feasible. This is the crisis Slide 8 documents.

6. The Problem: Standard Gibbs Mixes Poorly

[SLIDE: Slide 8]

The diagnostic: Inefficiency Factor (IF)

> IF $= k$ means you need k correlated MCMC draws to match the statistical accuracy of *one* independent draw.
>
> IF $= 5000$ with $M = 100{,}000$ draws $\Rightarrow$ effective sample size $= 20$. Practically useless.

Finance analogy. Estimating the average daily return of a strategy by sampling five consecutive crisis days that all look alike. Those five observations contain far less information than five draws from genuinely different market regimes. IF quantifies exactly this redundancy.

The root cause: a geometric problem

$\mathbf{\Lambda}$ and $\mathbf{f}_t$ lie on a **narrow posterior ridge**. Large λ_{jj} requires small f_{jt} to keep $\mathbf{\Lambda f}_t$ constant — and vice versa. Sequential Gibbs updates cannot break this dependency: every update of λ_{jj} is immediately compensated by the next update of f_{jt}. The chain inch-worms along the ridge, barely moving.

Finance illustration. You are trying to separately estimate "how strong was the dollar trend today" and "how sensitive is EUR/USD to the dollar trend". These lie on a ridge of observational equivalence. The sampler shuffles endlessly: strong trend with moderate sensitivity, moderate trend with strong sensitivity. It never settles.

The simulation evidence

Setup: $m = 10$, $r = 2$, $T = 1000$, 100 datasets, 5.1 million iterations each.

	Avg. IF for λ_{11}	**Eff. sample size (from 100k draws)**
No interweaving	> 1000	< 100
Shallow ASIS	≈ 200	≈ 500
Deep ASIS	≈ 5	$\approx 20{,}000$

The most troubling aspect is not the average — it is the **variance across datasets**. IFs

range from 100 to over 5000 across replications. You cannot predict in advance how badly your sampler will perform. The sampler fails worst precisely when factor structure is strongest and loadings are largest — the economically most interesting cases.

> **Key point to stress.** A correct model with a poorly mixing sampler is useless. You have beautiful Bayesian theory, but if the MCMC chain never explores the posterior, your credible intervals are meaningless. This motivates everything that follows.

7. The ASIS Innovation

[SLIDE: Slides 9 & 10]

The core insight (Yu & Meng 2011)

Two **mathematically equivalent parameterisations** of the same model are often each efficient in different regions of the parameter space. Switching between them at every iteration combines the best of both worlds. No new model. No tuning parameters. Just a change of coordinates at each step.

Finance analogy. A portfolio analyst switches between describing positions in terms of individual stock weights and factor exposures. Both descriptions contain identical information. Some risk calculations are easier in stock space, others in factor space. ASIS does this automatically, at every MCMC iteration.

The two parameterisations

	Baseline	**Alternative**
Loading	λ_{jj} *free*	$\lambda^*_{jj} = 1$ fixed
Factor variance level	$\mu_{m+j} = 0$ fixed	μ_{m+j} *free*
Observation eq.	$\mathbf{y}_t = \mathbf{\Lambda}\mathbf{f}_t + \boldsymbol{\varepsilon}_t$	$\mathbf{y}_t = \mathbf{\Lambda}^*\mathbf{f}^*_t + \boldsymbol{\varepsilon}_t$

The link is a simple change of variables:

$$\mathbf{f}^*_t = \lambda_{jj}\,\mathbf{f}_t, \qquad \mathbf{\Lambda}^* = \lambda_{jj}^{-1}\,\mathbf{\Lambda}$$

The product $\mathbf{\Lambda}\mathbf{f}_t = \mathbf{\Lambda}^*\mathbf{f}^*_t$ is *identical*. The model is unchanged; only the coordinate system differs.

Concrete illustration. Today the dollar factor has loading 2 on EUR/USD and realisation 0.5. In the alternative parameterisation: loading 1, realisation 1.0. Same EUR/USD contribution, different coordinates.

Shallow interweaving — the first fix

Procedure:

1. Run standard Gibbs steps (a)–(c) in the baseline parameterisation.
2. **Transform:** $\mathbf{f}^*_t = \lambda_{jj}\,\mathbf{f}_t$.
3. **Resample** λ^2_{jj} from its conjugate posterior in the alternative parameterisation. This turns out to be a Generalised Inverse Gaussian (GIG) — closed form, no tuning (see

Appendix A1):

$$\lambda_{jj}^2 \mid \cdot \;\sim\; \mathrm{GIG}(p,\, a,\, b)$$

4. **Transform back:** $\mathbf{f}_t = (\lambda_{jj}^{\text{new}})^{-1}\, \mathbf{f}_t^*$.

By resampling λ_{jj} in the alternative parameterisation, where it has cleaner posterior geometry, the sampler makes **larger jumps** along the ridge. **Gain:** $\approx$ 2–8×**.**

Why is this not enough? The log-volatility path $h_{m+j,\bullet}$ is left frozen. But the volatility path was estimated conditional on the *old* loading scale. After resampling λ_{jj}, the volatility path is internally inconsistent with the new scale. The chain must then laboriously re-estimate the entire volatility path to catch up — and the slow mixing between λ_{jj} and $h_{m+j,\bullet}$ persists.

Deep interweaving — the full fix

Deep interweaving additionally transforms the **log-volatility path itself** when switching parameterisations.

The log-volatility shift: The scaled factor $f_{jt}^* = \lambda_{jj} f_{jt}$ has variance:

$$\mathrm{Var}(f_{jt}^*) \;=\; \lambda_{jj}^2\, e^{h_{m+j,t}} \;=\; \exp\underbrace{\big(\log \lambda_{jj}^2 + h_{m+j,t}\big)}_{=:\, h_{m+j,t}^*}$$

So the transformed log-variance is:

$$h_{m+j,t}^* \;=\; h_{m+j,t} + 2\log|\lambda_{jj}|,$$

and the level of the new AR(1) is $\mu_{m+j} = \log \lambda_{jj}^2$ — which is now *free*. The scaling has been fully absorbed into the SV level.

Procedure:

1. Run standard Gibbs steps (a)–(c).
2. **Transform:** $\mathbf{f}_t^* = \lambda_{jj}\, \mathbf{f}_t$ and $h_{m+j,t}^* = h_{m+j,t} + 2\log|\lambda_{jj}|$.
3. **Resample** λ_{jj} via independence Metropolis–Hastings, using a Gaussian approximation to the conditional likelihood. No tuning required.
4. **Transform back:** $h_{m+j,t}^{\text{new}} = h_{m+j,t}^* - 2\log|\lambda_{jj}^{\text{new}}|$ and $\mathbf{\Lambda}^{\text{new}} = \frac{\lambda_{jj}^{\text{new}}}{\lambda_{jj}^{\text{old}}}\mathbf{\Lambda}^{\text{old}}$.

Gain: up to 400×**.**

Why deep beats shallow

Shallow breaks the $(\lambda_{jj},\, \mathbf{f}_t)$ dependency but leaves the $(\lambda_{jj},\, h_{m+j,\bullet})$ dependency intact. Deep breaks **both** simultaneously. After resampling λ_{jj}, the volatility path is immediately recalibrated to the new scale.

Finance analogy. Shallow interweaving is like updating your currency factor exposures but forgetting to recalibrate the factor volatility estimates. Your VaR calculation is now internally inconsistent — loadings suggest one scale, volatilities suggest another. Deep interweaving recalibrates everything simultaneously, keeping the model coherent at every step.

> **Likely question:** "Is ASIS just a trick specific to this model?"
> No. ASIS is a general principle from Yu & Meng (2011): whenever you have two equivalent parameterisations that are each efficient in different regions, switching between them at each iteration is always valid and often dramatically improves mixing. It applies broadly — factor models, random effects models, latent variable models of all kinds.

> **Likely question:** "Does the MH step in deep interweaving require tuning?"
> No. The Gaussian approximation to the conditional likelihood is constructed analytically by matching the mean and variance of $p(h^*_{m+j,1}, \ldots, h^*_{m+j,T} \mid \mu_{m+j}, \phi, \sigma^2)$, which is Gaussian in μ_{m+j}. The proposal is automatic.

8. Simulation Evidence

[SLIDE: Slide 11]

Setup: $m = 10$, $r = 2$, $T = 1000$, 100 replications, 5.1 million draws each.

	No interweaving	**Shallow**	**Deep**
λ_{11} (avg. IF)	> 1000	≈ 200	≈ 5
Median IF across $\mathbf{\Lambda}$	∼400–5000	∼50–400	∼3–20
Efficiency gain	—	2–8×	up to **400**×
Factor $f_{1,T}$ (IF)	high	moderate	low
Factor log-var $h_{m+1,T}$ (IF)	high	high	substantially reduced

Source: Tables 1–2, Kastner et al. (2017).

Key observations:

- Trace plots confirm **immediate mixing** with deep ASIS — the standard sampler barely moves over the same number of iterations (Figure 1 of the paper).
- The standard sampler's IFs vary catastrophically (100–5000+) across datasets; deep interweaving reduces this **uniformly** to single digits. Reliable and predictable.
- Shallow interweaving helps $\mathbf{\Lambda}$ and $\mathbf{f}$ but leaves the factor log-variance IF high — confirming that the $(\lambda_{jj}, h_{m+j,\bullet})$ dependency is the bottleneck that only deep interweaving resolves.
- Log predictive Bayes factors correctly recover $r = 2$ — model selection works reliably once the sampler actually mixes.

9. Application: 26 EUR Exchange Rates (2005–2015)

[SLIDE: Slide 12]

Setup: $m = 26$ daily EUR exchange rates, $T = 2649$ log-returns, $r = 4$ factors selected via log predictive Bayes factors, 500,000 post-burn-in draws. Data used *as-is*: includes the CHF spike (January 2015) and the RUB collapse (December 2014).

Four interpretable factors emerge from the data

- **F1 — Global USD/CNY factor.** The broad dollar trend. Almost every currency loads on this.
- **F2 — Eastern Europe.** ZAR, HUF, PLN move together; JPY loads *negatively* — it appreciates in risk-off environments when EM currencies fall.
- **F3 — Commodity currencies.** AUD, NZD, CAD — driven by global commodity prices.
- **F4 — Tiger Cub economies.** SGD and Asian peers.

These factors were not imposed — they **emerged from the data** entirely through the Bayesian inference procedure.

Key empirical findings

- Factor volatilities spike around the **2008 crisis** and again during 2014–15, capturing regime changes automatically.
- CNY and HKD show near-perfect correlation with USD throughout the sample — their loadings on F1 dominate, reflecting the de-facto dollar peg.
- The RUB–USD correlation **collapsed from** ≈ 0.9 **to** ≈ 0.4 during 2008–09 — visible directly in the time-varying posterior correlations.
- The CHF unpegging (January 2015) and RUB collapse (December 2014) are absorbed entirely by their *idiosyncratic* SV components. The global factor structure is not distorted by these country-specific shocks. This is the multivariate analogue of how ω_i absorbed outliers in Task 2.

> **Likely question:** "How do you choose $r = 4$?"
> Via log predictive Bayes factors — the same model selection framework discussed in Week 3. You compare the out-of-sample predictive likelihood of models with $r = 1, 2, 3, 4, 5, \ldots$ and select the one with highest log predictive score. The simulation study confirms this procedure correctly recovers the true r when the data-generating process is known.

10. Takeaways & Practical Outlook

[SLIDE: Slide 13]

Connections to the course

- **Data augmentation (Weeks 2–3).** The Factor SV Gibbs sampler is a direct generalisation of the ω_i augmentation from Task 2. Same philosophy: introduce latent variables to make conditional distributions tractable.
- **Model selection (Week 3).** Bayes factors select r — the same framework as discussed in the lecture.
- **Shrinkage (Week 3).** The Gaussian prior on $\mathbf{\Lambda}$ acts as a Ridge penalty, preventing overfitting in high dimensions.

- **Key lesson.** A correct model with a poorly mixing sampler is useless. ASIS is a general remedy that applies far beyond this specific model.

Three practical applications for the hedge fund

Portfolio optimisation. Replace the backward-looking sample covariance with the posterior mean of $\mathbf{\Sigma}_t$ at each annual rebalancing. Integrate out estimation uncertainty via posterior draws. The result: more stable Markowitz weights that do not flip sign between rebalancings, and a covariance estimate that is genuinely current rather than a historical average.

Risk management. The full predictive distribution of $\mathbf{y}_{T+1}$ delivers exact Bayesian Value-at-Risk and Expected Shortfall without Gaussian assumptions — capturing the fat tails and volatility clustering that standard models miss.

Correlation trading. Time-varying posterior correlations with full uncertainty bands signal *exactly when* stable relationships — like the RUB–USD co-movement before 2008 — are breaking down. You see the structural change as it happens, not six months after the fact.

> **Closing remark.** The R package `factorstochvol` (Hosszejni & Kastner) implements everything discussed in this talk, including ASIS, in efficient C++. It is production-ready and available on CRAN. For anyone wanting to apply this to real portfolio problems, this is the direct starting point.